

COMBINATORICS ASSIGNMENT 2

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6. Prove that

$$P(n, k) = k! \binom{n}{k}.$$

Proof. First let's consider how we choose our k -permutations of $[n]$. We begin by choosing the first element with n choices, the second with $n - 1$ choices, then all the way down to the k^{th} element with $n - k + 1$ choices, so we see that

$$P(n, k) = n(n - 1)(n - 2) \cdots (n - k + 1) = \frac{n!}{(n - k)!}.$$

On the right-hand side, we are choosing k -elements from $[n]$ in an unsorted fashion, but then we need to sort them in k ways which is $k!$, and we see that

$$k! \binom{n}{k} = k! \left(\frac{n!}{k!(n - k)!} \right) = \frac{n!}{(n - k)!}.$$

□

10. Prove that:

$$\sum_{k=0}^n r^{n-k} \binom{n}{k} = (r + 1)^n$$

Proof. We'll go ahead and consider the general case here for $(x + y)^n$. We want to prove:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

which if we multiply out, we get 2^n terms each of which can be arranged as $x^{n-k}y^k$ for $k \in [n]$. We obtain the coefficient for the $x^{n-k}y^k$ term by thinking about each of our 2^n terms written out without using exponents. That is, that each term is a string of length n of individual x 's and y 's (which are allowed to repeat). Now our goal is to count up the terms number of x 's is a specific value and the number of y 's is a specific value. Let the number of y 's appearing be k for convenience. Then we are choosing the number of k -subsets from $[n]$, which we know to be $\binom{n}{k}$. So we have

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

For the case in the problem we simply set $x = r$ and $y = 1$, and we are done.

Proof by induction of the Binomial Theorem. When $n = 0$, both sides equal 1, since $x^0 = 1$ and

$$\binom{0}{0} = 1.$$

Now suppose the equality holds for a given n ; we will prove it for $n + 1$. For $j, k \geq 0$, let $[f(x, y)]_{j,k}$ denote the coefficient of $x^j y^k$ in the polynomial $f(x, y)$. By the inductive hypothesis, $(x + y)^n$ is a polynomial in x and y such that

$$[(x + y)^n]_{j,k} = \begin{cases} \binom{n}{k}, & \text{if } j + k = n, \\ 0, & \text{otherwise.} \end{cases}$$

The identity

$$(x + y)^{n+1} = x(x + y)^n + y(x + y)^n$$

shows that $(x + y)^{n+1}$ is also a polynomial in x and y , and

$$[(x + y)^{n+1}]_{j,k} = [(x + y)^n]_{j-1,k} + [(x + y)^n]_{j,k-1}.$$

If $j + k = n + 1$, then $(j - 1) + k = n$ and $j + (k - 1) = n$. Hence the right-hand side is

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k},$$

by Pascal's identity.

If $j + k \neq n + 1$, then $(j - 1) + k \neq n$ and $j + (k - 1) \neq n$, so both terms are zero.

Thus,

$$(x + y)^{n+1} = \sum_{k=0}^{n+1} \binom{n+1}{k} x^{n+1-k} y^k,$$

which is the inductive hypothesis with $n + 1$ substituted for n , completing the inductive step. $\square$

16. Prove Vandermonde's Identity:

$$\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k}$$

Proof. The left-hand side counts all the k -element subsets of $[m + n]$. The right-hand side does the same thing according to the number of elements chosen from $[n]$. Indeed, we can first choose k elements from $[m]$ in $\binom{m}{k}$ ways, then choose the remaining $r - k$ elements from the set $\{m + 1, m + 2, \dots, m + n\}$ in $\binom{n}{r-k}$ ways. $\square$

19. Prove the identity:

$$\binom{n+1}{r+1} = \sum_{k=r}^n \binom{k}{r}$$

Proof. Let's go ahead and expand out the right hand side here so we see that:

$$\binom{n+1}{r+1} = \binom{r}{r} + \binom{r+1}{r} + \binom{r+2}{r} + \cdots + \binom{n}{r}$$

Now with regards to the identity in question, the left-hand side clearly counts all the $r+1$ -element subsets of $[n+1]$. And if we separate the right-hand side into cases according to the largest element. That is, there are $\binom{r}{r}$ subsets of $[n+1]$ that have $r+1$ elements whos largest element is $r+1$; there are $\binom{r+1}{r}$ subsets of $[n+1]$ that have $r+1$ elements whos largest element is $r+2$ and so on. In general there are $\binom{r+k}{r}$ subsets of $[n+1]$ that have $r+1$ elements whos largest element is $r+k+1$ for all $k \leq n-r$. $\square$

20. Prove the identity:

$$2^n - 1 = \sum_{k=0}^{n-1} 2^k$$

Proof. We begin by noticing that the left hand side counts the non-empty subsets of $[n]$. To see the count of the right-hand side we count by classifying all the subsets of $A \subseteq [n]$ by each's largest element $m \in A$ where $m \in \{1, 2, \dots, n\}$, $\max(A) = m$. We notice that every element of A is chosen from $[n]$, and elements $\{1, 2, \dots, m-1\}$ can be included or not independently. So we have that:

$$2^n - 1 = \sum_{m=1}^n 2^{m-1}.$$

A simple shift in indices $k = m - 1$ then results in:

$$2^n - 1 = \sum_{k=0}^{n-1} 2^k.$$

$\square$

22. Prove the identity:

$$\binom{n}{k} \binom{k}{t} = \binom{n}{t} \binom{n-t}{k-t}$$

Proof. The left-hand side is the number of ways that we can choose S of k -elements from $[n]$, and then subsequently choose t -elements from S . Regarding counting the right-hand side we see that we are immediately choosing t -elements from our original set of $[n]$. But then we are left with a set of $[n-t]$ from which we need to choose $k-t$ elements. So it works. $\square$

24. Prove the identity:

$$\sum_{k=0}^m \binom{m}{k} \binom{n}{r+k} = \binom{m+n}{m+r}$$

Proof. Let A and B be disjoint sets with $|A| = m$ and $|B| = n$. We want to find the number of ways to chose $S \subseteq A \cup B$ with $|S| = m+r$, clearly $\binom{m+n}{m+r}$.

Now to find the right-hand side we count by splitting according to how many are chosen from A . For any such subset S , let $k = |S \cap A|$. We see the following

- $j \in \{0, \dots, m\}$, and
- once j is fixed we must choose:
 - j -elements from A : $\binom{m}{j}$ ways, and
 - the remaining $((m+r) - j)$ -elements from B in $\binom{n}{(m+r)-j}$ ways.

So the total count follows as:

$$\sum_{j=0}^m \binom{m}{j} \binom{n}{(m+n)-j}$$

Now we re-index setting $k = m - j$ yielding $(m+n) - j = r + k$ with $\binom{m}{j} = \binom{m}{m-j} = \binom{m}{k}$, and sum becomes:

$$\sum_{k=0}^m \binom{m}{k} \binom{n}{m+k}$$

and we are done. □

26. Prove the identity:

$$\sum_{k=0}^n k^2 \binom{n}{k} = n(n-1)2^{n-2} + n2^{n-1} = n(n+1)2^{n-2}$$

Proof. Begin with the binomial theorem for $(x+1)^n$, which follows as:

$$(x+1)^n = \sum_{k=0}^n \binom{n}{k} x^k$$

Now differentiate both sides, and we get

$$n(x+1)^{n-1} = \sum_{k=1}^n k \binom{n}{k} x^{k-1}$$

Now multiply both sides of the equation by x yielding:

$$nx(1+x)^{n-1} = \sum_{k=1}^n k \binom{n}{k} x^k$$

Differentiating again we get

$$n[(x+1)^{n-1} + (n-1)x(1+x)^{n-2}] = \sum_{k=1}^n k^2 \binom{n}{k} x^{k-1}$$

Now if we substitute in $x = 1$ we get

$$\begin{aligned}n[2^{n-1} + (n-1)x2^{n-2}] &= \sum_{k=1}^n k^2 \binom{n}{k} \\n(n+1)2^{n-2} &= \sum_{k=1}^n k^2 \binom{n}{k}\end{aligned}$$

□

33. Prove the identity:

$$\sum_{k=n-s}^{m-r} \binom{m-k}{r} \binom{n+k}{s} = \binom{m+n+1}{r+s+1}$$

Solution.

□